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NeuroAID: An explainable artificial intelligence (XAI)-based brain tumor detection system using MRI analysis

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SUMMARY

Brain tumors are the fifth most common cancer worldwide, with 400,000 new cases diagnosed annually and a survival rate of 34%. Early detection using Magnetic Resonance Imaging (MRI) screening is critical, as tumor size can double in 20 days. In underdeveloped regions, radiologist shortages lead to costly, delayed, and error-prone diagnoses, averaging a 4% misdiagnosis rate. Most artificial intelligence (AI) models for MRI analysis in literature rely on black-box architectures that lack interpretability. We hypothesized that a deep learning model with a substantial number of parameters (e.g., ResNet50) would outperform an object detection algorithm (e.g., VGG16) for screening tumors, and that the pixel-based methods (e.g., XGBoost) would provide reliable classification of tumor types due to the spatial relevance of tumor location. To test these hypotheses, we developed a novel hybrid algorithm that integrates transfer learning–based features with machine learning to enable robust tumor screening, minimizing false positives and negatives. We found that feature fusion and hyperparameter optimizations led to 98% accuracy and an F1 score, which is a measure of predictive performance, of 0.98, which surpasses reported benchmarks in prior studies. For classification, XGBoost performed the best with 94% accuracy and F1 scores above 88% for all tumor types. We then developed feature-importance and saliency maps to provide explainable neurological insights. We deployed the best-in-class models in a secure health application, NeuroAID: Neurological AI-enabled Diagnostics, allowing neurologists to upload MRIs and obtain interpretable results instantly. Our findings help expand access to brain tumor screening in rural and underdeveloped countries.

INTRODUCTION

Brain tumors are abnormal growths of cells within the brain or central nervous system that pose significant threats to patient health by compressing nearby structures and disrupting essential neurological functions (1,2). In 2020 alone, a quarter of a million fatalities were reported worldwide due to cancerous brain tumors (3). Tumors are categorized according to the type of brain tissues from which they originate. For example, gliomas form in the glial cells, and meningiomas start in the spinal cord or protective lining of the brain (2,4). Another way of classifying tumors is malignancy—benign tumors typically do not spread to other tissues, while cancerous tumors can spread quickly, resulting in a low survival rate (2,5).

Detecting brain tumors in their incipient stages is crucial for devising personalized treatment plans and improving the chances of survival. Magnetic resonance imaging (MRI), a non-invasive imaging technique that avoids ionizing radiation, plays a crucial role in brain tumor diagnosis. MRI provides high-resolution images that allow clinicians to assess tumor location and size, enabling the development of tailored treatment plans (6–8). MRI is also useful for tracking the progress of a brain tumor over time and the effectiveness of treatment based on periodic scans (9). However, the diagnostic process often depends heavily on manual interpretation of MRI scans, which can be inconsistent due to human error and variability in training and experience (6). While MRI can detect abnormalities, it may not always determine whether the detected tumor is cancerous, and confirmation of malignancy typically requires biopsy—a costly and invasive procedure (7). In regions with limited healthcare infrastructure, these diagnostic steps are further delayed by a shortage of radiologists, leading to longer wait times and increased risk of misdiagnosis (10,11). Brain tumors are commonly misdiagnosed, affecting the health of thousands of patients and costing millions of dollars (12). As a result, there is growing interest in using automated techniques to aid in the early detection of tumors. A robust approach to tumor identification and classification is crucial for identifying explainable features — that is, insights that clearly show which features or patterns the model used to make its prediction, allowing clinicians to understand and trust the outcome (6–8).

In the past decade, machine learning (ML) and deep learning (DL) approaches have proven to be powerful tools for advancing medical diagnosis. These artificial intelligence (AI) techniques are trained on large volumes of data to identify features and patterns in the images, resulting in improved accuracy of screening diagnosis (1,12,13). AI-enabled medical diagnostics are a lifesaver for countries with underdeveloped healthcare systems. In places like Africa, South America, and parts of Asia with a lower density of medical doctors, ML can provide a way to diagnose people that would be impossible otherwise (12). This technology could also benefit developed healthcare systems facing staff shortages and backlogs (e.g., the UK's National Health Service (NHS) crisis). DL has become increasingly popular for medical image detection and has been successfully used in the analysis of breast cancer and diagnosis of lung or skin cancer.

Success of ML models depend heavily on the availability of large volumes of training data. For medical image classification problems, when large datasets are not readily available, transfer leaning techniques have been employed to achieve higher prediction accuracy. Transfer learning uses a pre-trained convolutional neural network (CNN) model to improve performance on a related task (9,10).

At present, medical image diagnostic tasks are subjective in nature based on the experience level of the doctor and quality of the MRI scan. For doctors and patients alike, it is important to have AI-enabled diagnostics results accompanied by a sound explanation of why the model predicted a certain tumor class or what aspects of the image determine the selection of each class. Explainable artificial intelligence (XAI) is becoming a critical need in the medical field, as it provides transparency and clarity into how complex DL models arrived at their predictions (14,15). This enhances transparency in healthcare management, improves diagnostic quality, and drives accountability, equity, and higher ethical standards (15).

We proposed and evaluated feature-based deep transfer learning fusion techniques to determine their efficacy in providing a robust approach for brain tumor screening and classification. We investigated VGG16 and ResNet50 transfer learning models separately, as well as in combination, to improve the fidelity of the models (16,17). We selected ResNet50 for its deeper architecture suited to capture more complex spatial features, while VGG16 served as a strong baseline model due to its proven performance in image classification tasks. We considered a variety of different neural network model architectures for tumor detection and classification. This contrasts with the previous work based on pixel-based algorithms, like k-neighbors classifier, naïve bayes classifier, random forest, logistic regression, and XGBoost,(11). Unlike DL models such as VGG16 and ResNet50, which learn hierarchical spatial features through convolutional layers, pixel-based algorithms treat each pixel's intensity as an independent feature, without capturing spatial relationships between them (12).

We examined whether feature-based hybrid deep learning models could serve as reliable and effective tools for screening and classifying brain tumors—specifically glioma, meningioma, and pituitary tumors—from MRI scans. We first hypothesized that transfer learning models using features extracted from ResNet50—a deeper architecture with more parameters—would outperform VGG16 and pixel-based machine learning models for binary tumor screening. We also hypothesized that is that for the tumor classification problem, pixel-based machine models will perform better than the feature-based models because of the inherent classification of tumors being dependent on the tumor's location.

To evaluate these hypotheses, we implemented and evaluated various hybrid detection and classification pipelines using standardized MRI datasets based on F1 score which minimizes the false positives and false negatives (18). Our results demonstrate that the hybrid ResNet50+K-neighbors classifier (KNC) model achieved strong detection performance, and XGBoost provided robust classification across tumor types, supporting both hypotheses. We also generated saliency maps and feature importance visualizations to improve interpretability. Finally, we deployed the highest-performing models in a secure, clinician-facing web application, NeuroAID, with the goal of enabling fast, explainable diagnostics, even in low-resource settings.

RESULTS

, We developed a three-stage explainable brain tumor diagnostic framework, by combining best-in-class detection models with second-stage classification modelsand identification keyfeatures influencing the prediction in the third stage (**Figure 1**).

Binary detection and multi-class classification

We developed and validated clustering-based binary classification models (tumor vs. no tumor) as an initial diagnostic tool to enable reliable brain tumor detection. We proposed a unique approach in this research, which combines the power of feature extraction capabilities of one or more transfer learning models like VGG16 and uses those features as the inputs to the ML model like KNC. Once our initial model detected a tumor, we applied a separate image-classification model in the second step to differentiate between three types of brain tumors—glioma, meningioma, and pituitary tumors—using brain MRIs to assist physicians in detecting tumors in initial stages with high accuracy (**Figure 1**) (19).

Hyperparameter optimization

We performed parametric studies on key hyperparameters for various algorithms over the corresponding range of values/settings to optimize the performance of these models (**Table 1**). Towards the goal of minimizing the false positives and false negatives, we selected F1 score as the key criterion to maximize. Model performance metrics measured in terms of accuracy, precision, and F1 score were optimized by conducting sensitivity studies in the hyperparameter space.(**Figure 2**). For instance, the essential training parameter for the XGBoost algorithm was the learning rate. The accuracy of the models changed with learning rates ranging from 0.01 to 1, with the optimal learning rate identified as 0.41 (**Figure 2**). We conducted parametric sensitivity analysis on the random forest classifier, focusing on the number of estimators. For DL and transfer learning models, we varied multiple parameters, including the number of epochs and optimizer types, to identify optimal configurations. Our sensitivity analysis of accuracy and loss across different epoch values for the VGG16 model identified five epochs as the optimal setting (Figure 2).

Model performance metrics

The most common metric for classification model is prediction accuracy; however, final accuracy is most reliable when the dataset contains an equal number of samples from each class. Due to the dataset's class imbalance, we evaluated performance using metrics beyond accuracy. For each algorithm, we calculated precision, recall, and F1 score from the corresponding confusion matrix (**Table 2**). The goal was to achieve a high F1 score for each of the classes.

The KNC with ResNet50 features (KNCR algorithm) turned out to be the best detection algorithm, giving a remarkably high F1 score of 0.98 on the test data as well as the highest accuracy of 0.98 (**Table 2)**. The K-neighbor clustering using the VGG16 model for feature extraction (KNCV algorithm) performed comparably using a significantly smaller set of features than ResNet50 ($p < 0.05$, two-sample t-test on feature vector dimensions). This approach can be appropriate in scenarios with limited computational resources. The data supports our first hypothesis, which posits that models leveraging ResNet50-extracted features would outperform VGG16 and pixel-based models in tumor screening, as demonstrated by the KNCR model's superior F1 scores.

It was interesting to find that the XGBoost algorithm performed best with an F1 score of at least 0.97 and overall accuracy of 0.94 (**Table 2**). For the classification models, there are separate metrics for each of the tumor classes. A robust model consistently achieves high F1 scores across all classes. With the balanced nature of DS2, it was encouraging to observe that XGBoost performed uniformly well in classifying all tumor types. The KNCR model worked equally well, except for with glioma which had an F1 score on the lower side. Superior performance of the XGBoost model for classification supported our second hypothesis that the pixel-based algorithms which leverage spatial features perform better in classification of tumor types.

We recorded the training and testing times per epoch for all models. While computational time for detection tasks remained under 2 minutes, training times for classification using DL and transfer learning algorithms ranged from 15 minutes to one hour in the GPU environment.

Model interpretability and explainability

We assessed model-level explainability for the ML models by generating feature importance plots from the trained models. These plots, presented as Pareto charts, highlight the most influential pixels contributing to tumor detection (Figure 3). These top 10 pixels provide extremely useful information regarding the presence of tumors in the MRI scan, as they correspond to regions where the model detects significant deviations in intensity or texture that are commonly associated with tumor tissue. The feature importance plot was based on the correlation of the pixel intensity with the class differentiation at the overall model level. Another way to look at the feature importance is to generate a heat map overlay on the MRI scan. This shows the regions of the brain where the AI algorithm focused to detect the tumor. It can be thought of as similar to the visual diagnostic process employed by skilled radiologists, who interpret abnormalities based on nuanced patterns in intensity and texture. For CNN and DL models, it is possible to extract important pixels at the specific scan level, which is especially useful for a patient to know in terms of what AI algorithm saw to classify their scan into a particular class. SHapley Additive exPlanations (SHAP) plots present the relationship between key features and their contribution to the model's prediction. For a specific patient MRI, high intensity pixels on a heatmap are indicative of regions where the algorithms found abnormal growth of cells as captured in the MRI (**Figure 3**).

Web application deployment: Neuro*AID*

As a final product of this study, we deployed the best-performing detection and classification models in a secure, easy-to-use web-based diagnostic application called Neuro*AID* (Neurological AI-enabled Diagnostics). The interface lets users upload an MRI, pick detection or classification, and see real-time predictions with probabilities—plus optional metrics, confusion matrices, and interpretability visuals (saliency maps, SHAP). (**Figure 4a**). This platform illustrates accessible, explainable AI–based screening support for clinical use, and we validated it on multiple samples from the held-out dataset (**Figure 4b**).

By using XAI, medical professionals can understand the rationale behind the model predictions and apply AI technology with clarity and confidence. Unlike traditional "black box" models, explainable AI offers transparency into which features contribute to a decision, helping clinicians verify, trust, or even challenge the model's suggestions. While radiologists rely on years of experience and visual assessment, XAI provides complementary insights by consistently analyzing complex patterns that may be difficult to detect by the human eye. This not only improves diagnostic confidence but also supports faster decision-making and more personalized patient care (**Figure 5**).

DISCUSSION

We applied novel 3-stage explainable framework to test the hypothesis that transfer-learning features from ResNet-50 outperform VGG-16 and pixel-based baselines for binary tumor screening. Also, the secondary hypothesis for classification was that the pixel-based models outperform feature-based models because class identity depends strongly on tumor location.

Following the evaluations of a variety of ML/DL algorithms for their ability to detect and classify brain tumors from MRI images, the multi-class brain tumor detection model based on features extracted from transfer learning models used as inputs to the K-neighbor algorithm was developed to achieve robust, high-fidelity predictions, with accuracy of 0.98 as well as F1 score of 0.98. This hybrid algorithm worked well for classifying tumors as well. However, XGBoost was found to be the best model for classifying brain tumors in terms of achieving the highest overall accuracy of 0.94 and high F1 scores for all three target classes. Data supported the original hypothesis that the transfer-learning based CNN classification models built on the training data could effectively classify the tumors into four classes representing no tumors, glioma, meningioma, and pituitary gland tumors. There is an opportunity to further improve robustness by combining the features from multiple transfer learning models. Although the models achieved high F1 scores, we did not perform a direct human-AI comparison against radiologist assessments. Therefore, while these results are promising, it would be premature to conclude that the model outperforms clinicians. Future research could involve a controlled study in which expert radiologists and the AI system independently evaluate the same set of MRI scans. Such comparisons would provide valuable insight into the complementary strengths of human expertise and ML approaches in clinical workflows.

, We used explainable aspects of our ML/DL models to better understand the physical features (size and location) of brain tumors on MRI scans. We demonstrated feature importance plots at the model level and saliency maps at the patient level as effective tools for interpreting spatial features in MRI scans that drive tumor class predictions.

While prior studies have applied DL to brain tumor classification, they often rely solely on end-to-end CNN architecture and focus primarily on accuracy metrics (14). Our study introduces a novel hybrid approach that combines features extracted from deep convolutional networks (VGG16, ResNet50) with classical ML classifiers such as KNC and XGBoost. This feature-based fusion enables flexibility, interpretability, and enhanced performance in certain settings, such as with small or imbalanced datasets and limited computational resources (22). Additionally, we integrate model- and patient-level explainability through saliency maps, SHAP plots, and cohort heatmaps, addressing a critical gap in clinical trust and transparency. This not only supports clinician buy-in but also facilitates error analysis, bias detection, and more informed decision-making. Finally, by deploying the trained model through an accessible web interface (NeuroAID), we translate our methods into a first of its kind usable diagnostic tool for research and educational purposes.

We designed Neuro*AID* as a lightweight, browser-accessible application that enables rapid, interpretable brain tumor screening using 2D MRI scans (31). Its real-world implementation is particularly suited to low-resource settings, including rural hospitals or clinics without access to full-time radiologists. By offering an intuitive interface and real-time results, the tool can assist primary care providers and neurologists in making faster, more confident referral and treatment decisions. . Since its launch in February 2024, users across multiple regions, including the United States and South Asia, have tested the program, though formal adoption metrics (e.g., downloads, user counts) are still being collected. Going forward, we hope to expand the functionality of NeuroAID to support 3D scan integration and deploy multilingual versions to further support global health accessibility. It is important to state that NeuroAID has not been approved by any regulatory or medical authority. Additionally, while NeuroAID has been deployed as a publicly accessible platform, adoption metrics, long-term user engagement, and integration into clinical workflows remain to be studied.

The models developed in this study demonstrated robust performance; however, the study is not without limitations that warrant further investigation. First, although we achieved strong results on publicly available datasets, the generalizability of our models to real-world clinical data from diverse imaging centers remains to be evaluated. Future work should include validation on larger and more heterogeneous datasets, ideally using external cohorts. There may also be value in combining the two data sets used in this researchand developing one integrated model for potential improvements in the F1 score. Second, our models rely on 2D MRI slices, which, while computationally efficient, may miss contextual information available in full 3D volumes. There remains an opportunity to incorporate 3D CNNs or temporal models to better represent tumor shape and growth across slices. We can extend the classification model to incorporate additional tumor types, enabling the development of a more comprehensive diagnostic framework within the same system. To improve the confidence and explainability of the predictions, there is an opportunity to further extend the feature-extraction based models based on the physical attributes of the tumors (20). We can achieve this by integrating the spatial distribution and regions of high-intensity pixels into the analysis. Another limitation is that, although explainability tools such as saliency maps and SHAP plots provide insight into model decision-making, their alignment with clinical reasoning remains unvalidated due to the lack of systematic feedback from radiologists. Human-AI comparison studies, including clinician-in-the-loop evaluations, would strengthen trust and real-world applicability.

This work provides a strong foundation for future AI-driven diagnostic tools that prioritize not only accuracy but also interpretability, user accessibility, and clinical trust—critical components for real-world adoption.

MATERIALS AND METHODS

Data sets

The MRI datasets used in this study are publicly available on Kaggle (32) based on the MRI datasets collected in the medical studies (33,34). For the detection model training, Dataset 1 (DS1) containing the brain MRI images of 300 patients with randomized attributes on their age, lifestyle, and other health factors was used. MRI scans are inherently 3D volumes, composed of voxels (three-dimensional pixels) aligned along three anatomical planes. The DS1 provided 2D slices extracted from all three standard orientations: axial (horizontal, top-down), coronal (vertical, front-to-back, resembling a face view), and sagittal (vertical, left-to-right, or side view). Each of the images in the dataset is a gray scale 2D section of the brain with a resolution of [150x150] pixels. In terms of target variable, each image is labeled as "No Tumor" and "Tumor". Approximately 61.3% of patients in DS1 have tumors, and the remaining 38.7% of them do not have tumors (**Figure 6b**).

The second data set (DS2) used was an augmented and cleaned-up version of a Kaggle dataset with 7023 human brain 2D MRI slices classified into 4 target classes: glioma, meningioma, pituitary, and no tumor (32, 33, 34). A balanced DS2 was only used for multi-class classification. In this dataset, each of the MRI images is a gray scale 2D section of the brain with a resolution of [512x512] pixels (6). The MRI images used in this dataset have three different views including axial, coronal, and sagittal (19). In terms of target variable, each image is labeled as either "No tumor", "glioma", "meningioma", or "pituitary". This dataset is balanced with all the classes represented equally in both the training and testing partitions (**Figure 6c**). The testing dataset had 1311 images, and the training data had 5712 images. Only DS2 was used for training and evaluating the classification models, as it contains labeled examples of multiple tumor types and healthy controls. DS1 was used exclusively for the binary detection task.

Model training and testing

For model development and testing work, the Python language was selected for its ease of use and availability of open-source, functional, numerical, and statistical libraries, such as numpy, pandas, math, sklearn, keras, XgBoost as well as Matplotlib for visualization (23-29). Web development was done in the Streamlit environment (30).

Model development, execution and testing were performed in the Google Colab high RAM Python environment, version 3.10. We evaluated the performance of the model using four metrics: accuracy, precision, recall, and F1 score, based on data from the confusion matrix. Additionally, it is important for model training and predictions to be computationally efficient. Training times are a secondary variable.

The goal was to develop models that have high model accuracy but at the same time keep false positives and false negatives at a small number, which is critical for building a robust prediction. Therefore, the F1 Score, defined below, was the most important metric used in selecting the best model (Equation 1).

$$\text{F1 Sccore} = \frac{2 * \text{True Positives}}{(2 * \text{True Positives}) + \text{False Positives} + \text{False Negatives}} \qquad \text{(Equation 1)}$$

The data and analytics pipelines were implemented in Colab for the detection and classification tasks. All the image files have three channels, even though they are gray images. For the ML/DL models, the images were converted to one channel, and the size of 150x150 pixels was selected to satisfy the RAM requirements. Data preparation steps are key for building efficient ML/DL models. In the first step of image pre-processing and normalization, darker corners of the rectangular images were cropped out to focus all the images on the brain region of the MRI. Then, images were resized to 150x150 pixels and normalized to scale the intensity values between 0 and 1.. For the Transfer learning, all the three channels of the original images were used. For the DL algorithms, three channels were summarized into one channel. These images were flattened to 1 dimension of 22500 pixels for the ML algorithms. Next, target labeling was performed by changing the tumor type labels to numerical values by assigning 0 to no tumor, 1 to glioma, 2 to meningioma, and 3 to pituitary. The predictive power of the models is dependent on the size of the datasets, so horizontal reflections of the original images were performed to double the size of the datasets in the data augmentation step. Finally, the dataset was divided into 80% and 20% for training and testing, respectively, based on randomized sampling. As experimental control, model accuracy and precision achieved on the training partition was cross validated on the testing partition to achieve significant lift.

Several ML/DL algorithms were applied for the detection and classification stages, and hyperparameters were optimized for some of them to arrive at the most accurate model for that algorithm. One of the key steps in converging to the optimal training parameters for the various models was to perform a parametric study to see how the performance metrics change with respect to hyperparameters such as learning rate, number of estimators, batch size, epochs, etc. (**Figure 2**, **Table 1**). We chose hyperparameters for our transfer-learning models (VGG16 and ResNet50) based on empirical tuning and established benchmarks in medical image classification. We used a batch size of 32 to balance training efficiency and model stability.. For the detection model, which has two classes, it was a 2x2 matrix with the top right cell corresponding to the false positive, or type I error, and the bottom left cell being the false negative, or type II error. Type 1 error could lead to unnecessary medical treatments; on other hand, type II error could lead to delay in getting proper treatment. Thus, a robust screening model should minimize both types of errors, the off-diagonal elements. For the classification model, the confusion matrix was 4x4 with the off-diagonal elements representing the misclassification of the types of tumors. The best model was selected based on achieving the highest accuracy without compromising the F1 score. We used the best models for detection and classification to predict the classes for the validation data set, which was not used in the model training and cross-validation.

Model deployment and application development

For developing neurological insights, feature importance plots and heat maps were used to understand the model-level features that were associated with the target class(es). We developed saliency maps for patient-level explainability. By comparing these maps for the MRI with and without tumors, fundamental understanding can be developed about the tumor attributes. The NeuroAID web application (https://neuroaid.streamlit.app/) was developed using the Streamlit Python library, which supports seamless integration of ML models with interactive web interfaces (23). The app includes dropdown menus for selecting the diagnostic mode (detection or classification), model of interest, and optional visual outputs. Once the user uploads a 2D MRI slice, the selected model processes the image and returns a prediction with accompanying probability. The interface was evaluated on multiple samples from the validation dataset to ensure functionality and accuracy alignment with offline model results.

Code for the model development, training and testing is provided in the following github repo: https://github.com/TMRmehta/NeuroAID.

REFERENCES

1. Kumar, Sandeep, et al. "Brain Tumor Classification Using Deep Neural Network and Transfer Learning." *Brain Topography*, vol. 36, no. 3, May 2023, pp. 305-318, https://doi.org/10.1007/s10548-023-00953-0.
2. "Brain Tumors and Brain Cancer." *Johns Hopkins Medicine*. https://www.hopkinsmedicine.org/health/conditions-and-diseases/brain-tumor. Accessed 15 November 2023.
3. Miller, Kimberly D., et al. "Brain and other central nervous system tumor statistics." *CA: A Cancer Journal for Clinicians*, vol. 71, no. 5, Sep. 2021, pp. 381-406, https://doi.org/10.3322/caac.21693.
4. McFaline-Figueroa, J. Ricardo, et al. "Brain Tumors." *The American Journal of Medicine*, vol. 131, no. 8, Aug. 2018, pp. 874-882, https://doi.org/10.1016/j.amjmed.2017.12.039.
5. Ilic, Irena, et al. "International patterns and trends in the brain cancer incidence and mortality: An observational study based on the global burden of disease." *Heliyon*, vol. 9, no. 7, 13 Jul. 2023, https://doi.org/10.1016/j.heliyon.2023.e18222.
6. Kibriya, Hareem, et al. "A Novel and Effective Brain Tumor Classification Model Using Deep Feature Fusion and Famous ML Classifiers." *Computational Intelligence and Neuroscience*, vol. 2022, 26 Mar. 2022, https://doi.org/10.1155/2022/7897669.
7. Abd-Ellah, Mahmoud Khaled, et al. "A review on brain tumor diagnosis from MRI images: Practical implications, key achievements, and lessons learned." *Magnetic Resonance Imaging*, vol. 61, Sept. 2019, pp. 300-318, https://doi.org/10.1016/j.mri.2019.05.028.
8. Mabray, M., et al. "Modern brain tumor imaging." *Brain Tumor Research and Treatment*, vol. 3, no. 1, Apr. 2015, pp. 8-23, https://doi.org/10.14791/btrt.2015.3.1.8.
9. Cè, Maurizio, et al. "Artificial Intelligence in Brain Tumor Imaging: A Step toward Personalized Medicine." *Current Oncology,* vol. 30, no. 3, 22 Feb. 2023, pp. 2673-2701, https://doi.org/10.3390/curroncol30030203.

10. Haq, Amin Ul, et al. “DACBT: deep learning approach for classification of brain tumors using MRI data in IoT healthcare environment.” *Scientific Reports*, vol. 12, no.1, 12 Sep. 2022, https://doi.org/10.1038/s41598-022-19465-1.
11. Shah, Faisal M., et al. “Brain Tumor Segmentation Techniques on Medical Images – A Review.” *International Journal of Scientific & Engineering Research*, vol. 10, no. 2, Feb. 2019, https://doi.org/10.13140/RG.2.2.18685.00486.
12. Kaifi, Reham. “A Review of Recent Advances in Brain Tumor Diagnosis Based on AI-Based Classification.” *Diagnostics*, vol. 13, no.18, Sep. 2023, https://doi.org/10.3390/diagnostics13183007.
13. Parmar, Ankita, et al. “A survey on Brain Tumor detection using deep learning.” *International Journal of Advance Engineering and Research Development*, vol. 7, no. 4, Apr. 2020,
14. Khan, Md Saikat Islam, et al. “Accurate brain tumor detection using deep convolutional neural network.” *Computational and structural biotechnology journal*, vol. 20, Aug. 2022, pp. 4733-4745, https://doi.org/10.1016/j.csbj.2022.08.039.
15. Yan, F., et al. "An Explainable Brain Tumor Detection Framework for MRI Analysis." *Applied Sciences*, vol. 13, no. 6, 2023, https://doi.org/10.3390/app13063438.
16. Simonyan, K., et al. “Very Deep Convolutional Networks for Large-Scale Image Recognition.” *International Conference on Learning Representations*, Sep. 2014, pp. 1409-1556, https://doi.org/10.48550/arXiv.1409.1556 .
17. He, Kaiming, et al. “Deep Residual Learning for Image Recognition.” *IEEE Conference on Computer Vision and Pattern Recognition*, 2016, pp. 770-778, https://arxiv.org/abs/1512.03385 .
18. Salukhe, A., et al. “Brain Tumor Detection using Image Processing.” *International Journal for Research in Engineering Application & Management*, vol. 9, no. 12, Mar 2024, pp. 2454-9150, https://doi.org/10.35291/2454-9150.2024.0091.
19. Gumaei, A., et al. “A Hybrid Feature Extraction Method with Regularized Extreme Learning Machine for Brain Tumor Classification.” *IEEE Access*, 2019, pp. pp. 36266–36273, https://doi.org/10.1109/ACCESS.2019.2904145.
20. Esmaeili, M., et al. “Explainable Artificial Intelligence for Human-Machine Interaction in Brain Tumor Localization.” *Journal of Personalized Medicine*, vol. 11, no. 11, Nov. 2021, https://doi.org/10.3390/jpm11111213.
21. Walter, Fiona M et al. “Missed opportunities for diagnosing brain tumours in primary care: a qualitative study of patient experiences.” *The British Journal of General Practice: The*

Journal of the Royal College of General Practitioners, vol. 69,681 (2019): e224-e235, https://doi.org/10.3399/bjgp19X701861.

22. Chaudhary, Ashish, et al. "Hybrid Deep Learning Approach for Brain Tumor Classification Using MRI Images." *Neural Computing and Applications*, vol. 34, 2022, pp. 16043–16054. Springer, https://doi.org/10.1007/s00521-021-06871-2.
23. NumPy. Version 1.26.0, NumPy Developers, 2024, https://numpy.org/. Accessed 14 December 2023.
24. Pandas. Version 2.2.1, pandas Development Team, 2023, https://pandas.pydata.org/. Accessed 14 December 2023.
25. Math Module. Python Standard Library, Python Software Foundation, 2024, https://docs.python.org/3/library/math.html. Accessed 14 December 2023.
26. Scikit-learn. Version 1.4.2, Scikit-learn Developers, 2023, https://scikit-learn.org/stable/. Accessed 14 December 2023.
27. Keras. Version 3.0.5, Keras Team, 2023, https://keras.io/. Accessed 14 December 2023.
28. XGBoost. Version 2.0.3, XGBoost Developers, 2023, https://xgboost.readthedocs.io/. Accessed 14 December 2023.
29. Matplotlib. Version 3.9.0, Matplotlib Development Team, 2023, https://matplotlib.org/. Accessed 14 December 2023.
30. Streamlit. Version 1.24.1, Streamlit, 2024, https://streamlit.io/. Accessed 14 January 2024.
31. Mehta, Tushar "NeuroAID: Explainable brain-tumor screening," Web application, ver. 1.0, 2024. [Online]. Available: https://neuroaid.streamlit.app/
32. Nickparvar, Masoud. *Brain Tumor MRI Dataset*. Kaggle, 2022, https://www.kaggle.com/datasets/masoudnickparvar/brain-tumor-mri-dataset. Accessed 10 November 2023.
33. Cheng, J., et al. "Enhanced performance of brain tumor classification via tumor region augmentation and partition." *PLOS ONE,* 2015, 10(10), e0140381. https://doi.org/10.1371/journal.pone.0140381
34. Cheng, et al**.** "Retrieval of brain tumors by adaptive spatial pooling and partition." *PLOS ONE,* 2016, 11(6), e0157112. https://doi.org/10.1371/journal.pone.0157112

Figures and Figure Captions

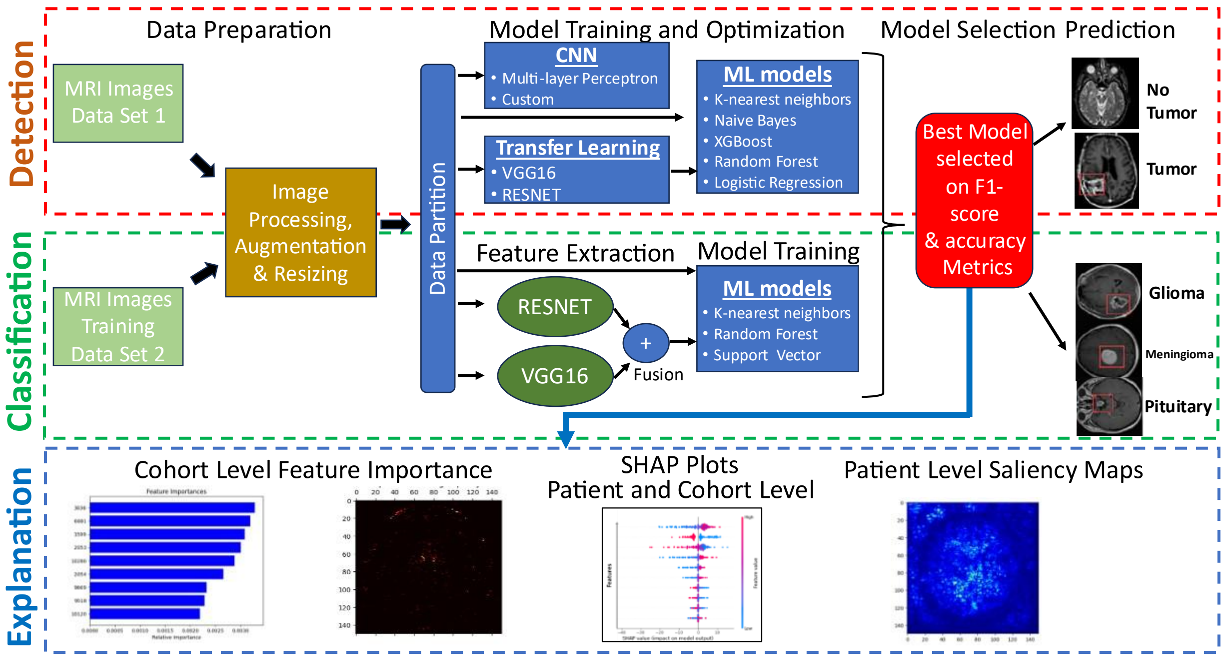

Figure 1. Overview of the three-stage workflow used in this study for brain tumor detection, classification, and explanation. (1) Detection: Identify whether a tumor exists. (2) Classification: Predict tumor type among glioma, meningioma, or pituitary. (3) Explanation: Generate patient- and cohort-level visual insights (saliency maps, feature importance plots). Models with different architectures were trained and compared to identify the best models, which can then be used to differentiate between tumor types and support neurologically meaningful insights in diagnostic applications.

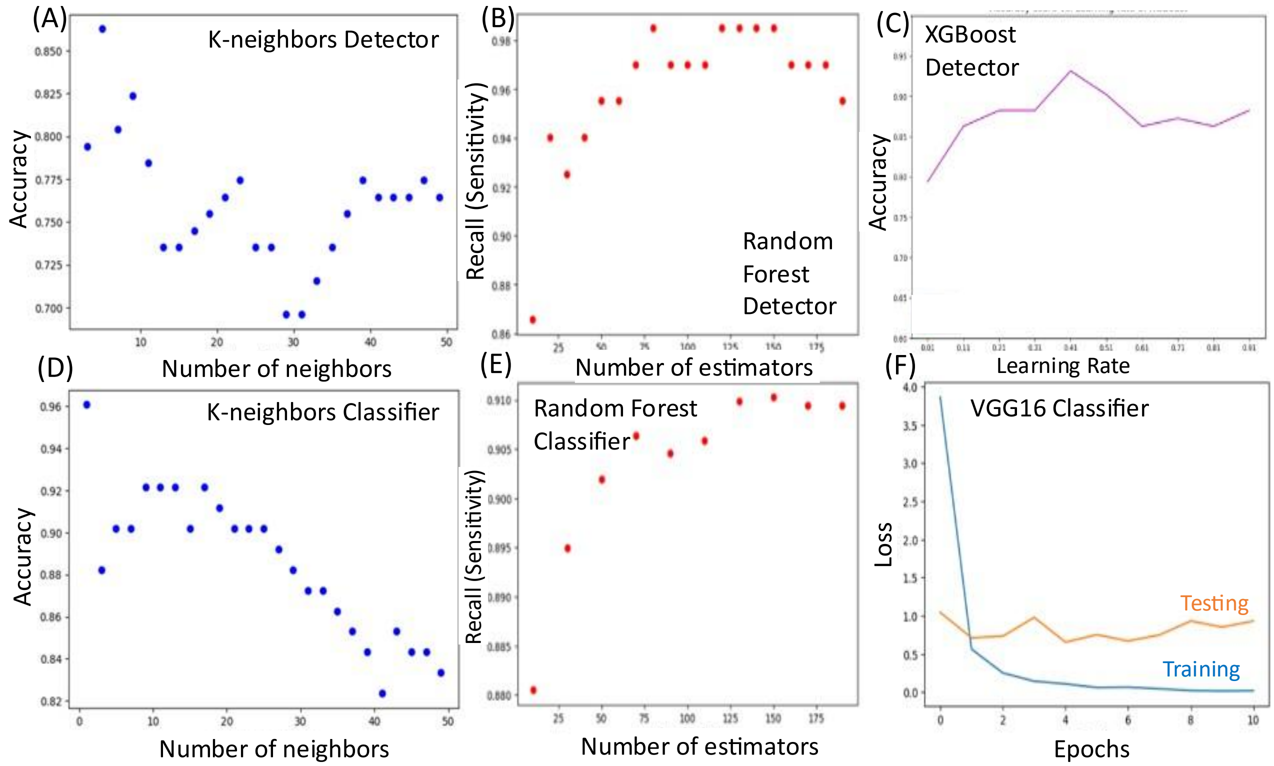

Figure 2. Parametric evaluation of hyperparameters across different detection and classification models. Optimal hyperparameter selection based on maximizing F1 score or minimizing loss for representative set of algorithms. **(A)** K-neighbors detector: Accuracy trends across varying neighbor counts; optimal performance observed at 5 neighbors **(B)** Random Forest detector: Sensitivity of detection plateaus near 125 estimators **(C)** XGBoost detector: Accuracy peaks at a learning rate of 0.41 **(D)** K-neighbors classifier: Highest accuracy achieved using 1 neighbor **(E)** Random Forest classifier: Peak sensitivity obtained with 150 estimators **(F)** VGG16 classifier: Training and testing loss curves indicate optimal learning around epoch 5.

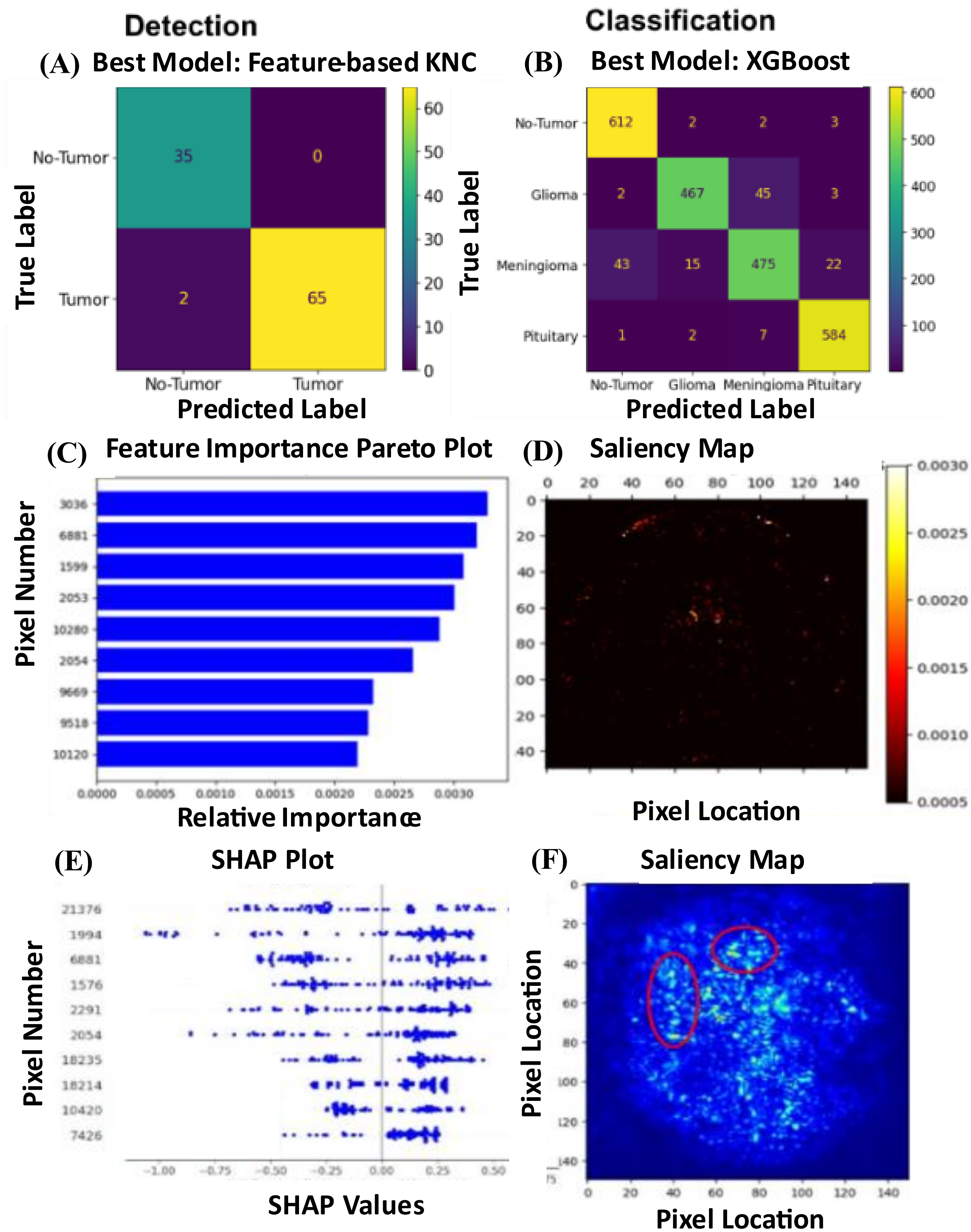

Figure 3. Model performance and explainability visualizations for brain tumor detection, and classification. (A) Confusion matrix for the best-performing detection model, ResNet50 features with KNC performed best with smallest number of off-diagonal elements. These

represent misclassifications where the predicted label does not match the actual label. **(B)** Confusion matrix for the best-performing classification model, XGBoost, which achieved the highest F1 scores across all four classes: no tumor, glioma, meningioma, and pituitary. **(C)** Cohort-level feature importance plot showing the top 10 most influential pixels used by the detection model to distinguish between tumor and no tumor cases. **(D)** Cohort-level heatmap highlighting the most predictive pixel regions across the standardized MRI dataset, illustrating common spatial features associated with tumors. **(E)** Patient-level SHapley Additive exPlanations (SHAP) dependence plot demonstrating how specific features influenced the classification model's prediction for an individual case. **(F)** Patient-level saliency map showing high-intensity pixels that contributed most to the model's prediction of tumor type, helping localize diagnostically important regions on the MRI scan.

(A) System architecture 3 -step screening diagnosis

(B) Best classification model (XGBoost) applied on validation dataset

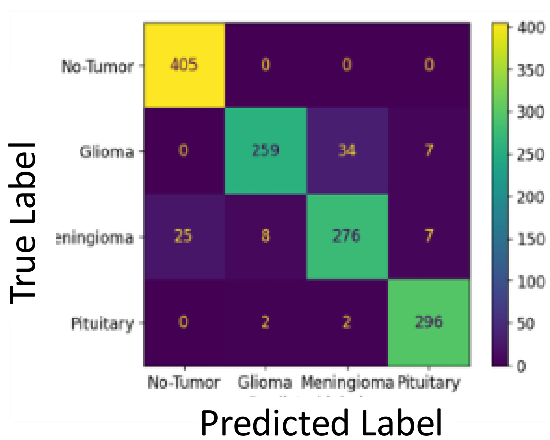

Figure 4. Deployment of the NeuroAID web application and independent field-trial performance of the best classification model. (A) System architecture for developing and deploying secure web app for detection and classification of brain tumors; screenshots of the app showing the diagnostic flow—image upload, model selection, output probabilities, and explainability visualizations. **(B)** Performance metrics for the independent validation data set using the best XGBoost classification model with overall accuracy of 94% and consistently high (>88%) F1 scores for all the four classes.

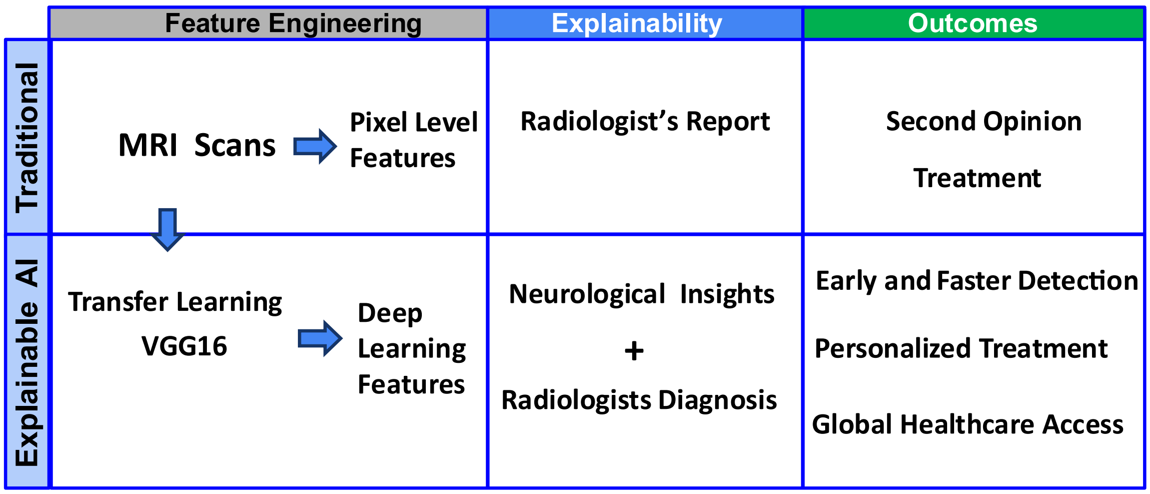

Figure 5. XAI-enabled MRI analysis as a lifesaver for patients in regions with underdeveloped healthcare systems. Explainable Artificial Intelligence (XAI) techniques integrate Convolutional Neural Networks (CNNs) and transfer learning models (e.g., VGG16) to analyze MRI scans, visualize results, and provide interpretable insights about structures and features in the MRI scans, giving neurologists confidence in recommending prompt personalized treatment for their patients.

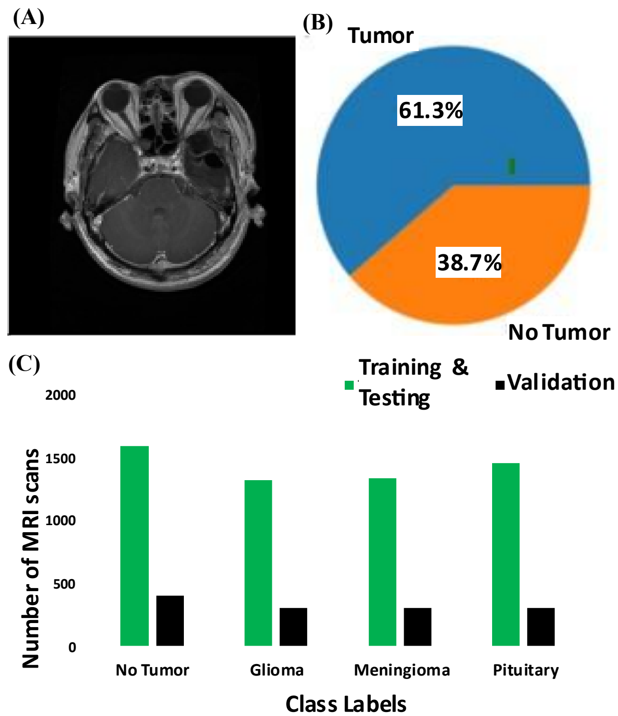

Figure 6. Summary of data sets used for training detection and classification models. (A): A typical MRI scan of the brain analyzed for detection and classification of tumors. **(B)** Pie chart summarizes Data Set 1 (DS1) with MRI images of 300 patients (sub-selected) with randomized attributes on their age, lifestyle, and other health factors used for training and testing binary detection models (32). **(C)**: Data Set 2 (DS2), which contained 7023 human brain MRI scans that were classified into 4 classes: no tumor, glioma, meningioma, and pituitary (32). DS2 was used exclusively for multi-class tumor classification. Images represent 2D slices from axial, coronal, and sagittal views. The bar chart shows class distributions in training/testing and validation partitions.

Tables with Captions

AI Algorithms	Detection	Classification	Hyperparameters	Range Settings
ML				
K-neighbors Classifier (KNC)	√	√	n-neighbors	[1,50]
Naive Bayes (NB)	√	√		
Logistic Regression (LR)	√			
XGBoost (XGB)	√	√	Learning Rate	[0.01,1]
Random Forest (RF)	√	√	Number of Estimators	[10,200]
DL				
Multi-Layer Perceptron (MLP)	√	√	Number of Layers Neurons per Layer	[1-5] [10,100,500]
CNN	√	√		
Transfer Learning				
VGG16	√	√	Optimizer Batch Size Epochs	Adam, SGD Various [1-50]
Feature-based ML				
VGG16+KNC (KNCV)	√	√		
ResNet50+KNC (KNCR)	√	√		
Feature Fusion-based ML				
(VGG16+Resnet50) + KNC (KNCF)	√	√		

Table 1: Design space. Summary of ML, DL, and transfer learning models evaluated for tumor detection and classification. For algorithms marked with green checks, corresponding hyperparameters and value ranges were tested. Black checks indicate the fixed parameters.

ML/DL Detection Algorithms	Image size	Accuracy	Precision			Recall			F1 Score		
K-neighbors Classifier (KNC), n =5	(*, 22500)	0.86	0.93			0.85			0.89		
Naive Bayes (NB)	(*, 22500)	0.69	0.89			0.6			0.71		
Logistic Regression (LR)	(*, 22500)	0.83	0.86			0.9			0.88		
XGBoost (XGB), lr=0.41	(*, 22500)	0.93	0.92			0.99			0.95		
Random Forest (RF)	(*, 22500)	0.94	0.93			0.99			0.96		
Convolutional Neural Net (CNN)	(*,150,150,1)	0.90	0.91			0.96			0.93		
Transfer Learning (VGG16)	(*,150,150,3)	0.94	0.96			0.96			0.96		
VGG16 Feature + KNC (KNCV)	(*,150,150,3)	0.96	1			0.94			0.97		
ResNet50 Feature + KNC (KNCR)	(*,150,150,3)	0.98	1			0.97			0.98		
(VGG16 +ResNet50) + KNC (KNCF)	(*,150,150,3)	0.97	1			0.96			0.98		
ML/DL Classification Algorithms			G	M	P	G	M	P	G	M	P
K-neighbors Classifier (KNC), n =1	(*, 22500)	0.93	0.87	0.92	0.96	0.91	0.86	0.98	0.89	0.89	0.97
Naive Bayes (NB)	(*, 22500)	0.62	0.5	0.39	0.79	0.83	0.33	0.63	0.62	0.36	0.7
XGBoost (XGB)	(*, 22500)	0.94	0.97	0.9	0.97	0.9	0.97	0.98	0.93	0.88	0.97
Random Forest (RF)	(*, 22500)	0.91	0.94	0.84	0.92	0.84	0.83	0.97	0.89	0.83	0.95
Transfer Learning (VGG16)	(*,150,150,3)	0.93	0.93	0.82	0.96	0.89	0.93	0.93	0.91	0.87	0.95
VGG16-Feature + KNC (KNCV)	(*,150,150,3)	0.93	0.91	0.88	0.93	0.88	0.86	0.98	0.89	0.87	0.95
ResNet50-Feature + KNC (KNCR)	(*,150,150,3)	0.93	0.87	0.89	0.86	0.86	0.86	0.99	0.87	0.88	0.97
(VGG16+ResNet50) + KNC (KNCF)	(*,150,150,3)	0.93	0.91	0.88	0.93	0.88	0.86	0.98	0.89	0.87	0.95

Table 2: Performance metrics. Top: Accuracy, precision, recall, and F1 score for binary detection task using DS1. Bottom: Per-class performance metrics (glioma, meningioma, pituitary) for multi-class classification using DS2. The best models achieved high F1 scores across all tumor types, with XGBoost and KNCR leading in performance. Highlighted rows represent best-performing model configurations selected for final deployment. Key: G, glioma; M, meningioma; P, pituitary tumor.